```
clear all
clc
close all
```

Properties

```
Ex = 23.2; Ey = 1.3; G = 0.90; % Msi
nu_x = 0.28;
nu_y = Ey / Ex * nu_x;

angles = [30 -45 60 0];
n_layers = length(angles);
h = 0.007 * ones(1,n_layers);

alphas = repmat([-6.8e-6; 32e-6; 0],1,n_layers);
betas = zeros(3,n_layers);

deltaT = (75-365) * ones(1,n_layers);
deltaC = zeros(1,n_layers);
```

Calculations using hw functions

```
Q = plane_stiffness(Ex,Ey,G,nu_x,nu_y)
Q_layers = repmat(Q,1,1,n_layers);
[A,B,D,alpha,beta,delta,Q_bar,Ts,z] = ...
    laminate_stiffness(Q_layers,angles,h);
A, B, D, alpha, beta, delta
Ts
Q_bar

% From homework
[ep_mid,k_mid,N_nm,M_nm,ep_free,ep_constrained,ep_residual,sigma_global,sigma
_local] = ...

nonmechanical_response(Q_bar,Ts,z,alpha,beta,delta,alphas,betas,deltaT,deltaC
);

N_nm * 1e6 % lb/in
M_nm * 1e6 % lb/in
ep_mid
k_mid
```

```
Q =

       23.302      0.36561            0
      0.36561       1.3057            0
            0            0          0.9


A =
```

```
      0.33279     0.098948      0.02818
     0.098948      0.17881      0.02818
      0.02818      0.02818      0.11391


B =

   0.00057995  -0.00031049  -0.00031374
  -0.00031049   4.1031e-05   0.00011648
  -0.00031374   0.00011648  -0.00031049


D =

   3.1026e-05   4.7259e-06   5.2331e-06
   4.7259e-06   5.8764e-06   2.2215e-06
   5.2331e-06   2.2215e-06   5.7035e-06


alpha =

       5.4503      -3.8637     -0.21624
      -3.8637       9.0955      -2.2544
     -0.21624      -2.2544       12.716


beta =

      -279.13       361.39        482.3
       289.15      -185.57      -714.05
       50.844      -639.83       928.97


delta =

        56425       -42908       -53550
       -42908    2.663e+05       -75515
       -53550       -75515   3.4557e+05


Ts(:,:,1) =

         0.75         0.25      0.86603
         0.25         0.75     -0.86603
     -0.43301      0.43301          0.5


Ts(:,:,2) =

          0.5          0.5           -1
          0.5          0.5            1
          0.5         -0.5   2.2204e-16
```

```
Ts(:,:,3) =

    0.25      0.75     0.86603
    0.75      0.25    -0.86603
 -0.43301   0.43301        -0.5


Ts(:,:,4) =

     1     0     0
     0     1     0
     0     0     1


Q_bar(:,:,1) =

   14.001    4.1675    6.9574
   4.1675     3.003    2.5674
   6.9574    2.5674    4.7019


Q_bar(:,:,2) =

   7.2348    5.4348   -5.4992
   5.4348    7.2348   -5.4992
  -5.4992   -5.4992    5.9692


Q_bar(:,:,3) =

    3.003    4.1675    2.5674
   4.1675    14.001    6.9574
   2.5674    6.9574    4.7019


Q_bar(:,:,4) =

   23.302   0.36561         0
  0.36561    1.3057         0
        0         0       0.9


ans =

   625.13
   247.44
   138.24


ans =

  0.66096
 -0.66096
```

```
     -0.48385


ep_mid =

    0.0017645
    0.0001829
     0.001072


k_mid =

    -0.004349
    -0.076288
      0.10055
```

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